Assignment 1 – Summary

GitHub repository

GitHub username	ishan-monie
Repository name	advanced-ml-hw1
Link to the code	https://github.com/ishan-monie/advanced-ml-hw1

All of the code used for this assignment is in this repository.

How to run the code

Step 1. Download the repository (green *Code* button, then *Download ZIP*) or clone it, and install the Python packages:

```
pip install numpy matplotlib
```

Step 2. Shuttle problem. Open a terminal in the `q1_shuttle` folder and run:

```
python run_q1.py     # solves the problem 3 ways and saves the 3 plots in figures/
python test_q1.py    # runs the 13 edge-case and correctness checks (PASS/FAIL for each)
```

The README in the repository repeats these instructions and explains what each file does. Instructions for the policy gradient questions will be added to the same README.

Question 2: Value and Policy Iteration (the shuttle problem)

Each period we choose whether to dispatch a shuttle (cost 100, takes up to 15 people) or wait (every person left waiting costs 2). Between 1 and 5 new people arrive each period. We solve this three ways. A lower cost is better in every graph.

(a) Enumeration, 500 periods

We start at the end of the 500 periods, where nothing is left to pay, and work backwards, keeping the cheaper of the two actions at every step. The graph shows the best expected total cost at time 0 for each number of people waiting.

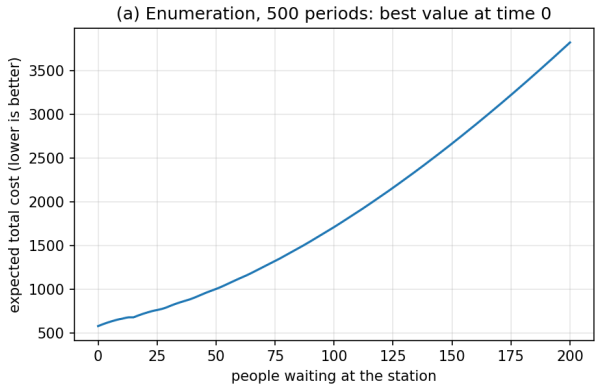


Figure 1. Enumeration: best expected cost at time 0 versus people waiting.

(b) Value iteration, no end date

The same step is repeated with no end date until the numbers stop changing (519 rounds). The curve is the same as in (a), which is expected: after 500 periods the far future is discounted to almost nothing, and the two differ by about 3 in a billion.

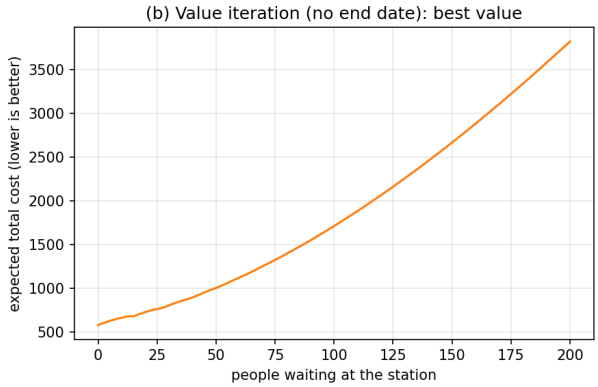


Figure 2. Value iteration: best expected cost versus people waiting.

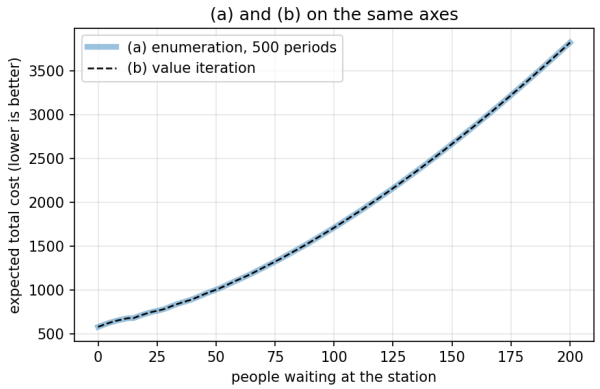


Figure 3. Figures 1 and 2 on the same axes. The two curves lie on top of each other.

(c) Policy iteration

Here we improve a plan instead of the costs. Starting from “never dispatch”, we measure how good the plan is, change it wherever another action is cheaper, and repeat. It settled after 6 rounds. The best plan is to wait while fewer than 13 people are waiting and to dispatch at 13 or more.

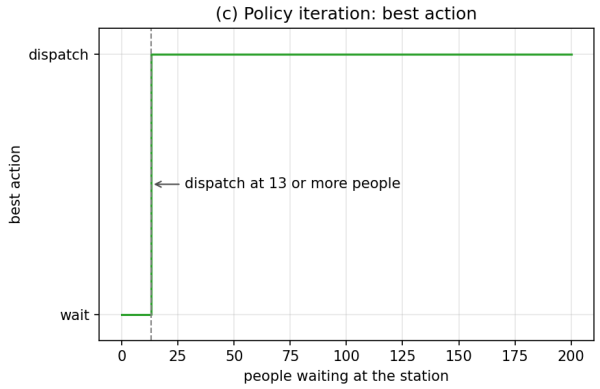


Figure 4. Policy iteration: best action versus people waiting.